

Appendix 4 - Utility Capacity Assessment Calculations

		CALCULATION SHEET	
PROJECT NO: 300291		CALC NO: 300291-PR-CA-0001	
TITLE: Asymchem PDF Extension Facility		ISSUE No: A1	
SUBJECT: Utility Capacity Assessment Calculations		BY: ME	
		CHKD: MC	
		APPROVED ME	
		DATE: 21-May-26	
Objective • This assessment determines whether existing utility systems can accommodate the new extension facility, or whether new/upgraded infrastructure is required. • The extension introduces 10 new reactors (R25–R34) ranging from 2,500L to 8,000L — significantly larger than the existing facility maximum of 2,500L (R12, R17). The extension also has a dedicated hydrogenation reactor (R35; 2500L compared to existing 250L R23 hydrog reactor) and 3x filter dryers (1x 1130L and 2x 1530L) • Detailed utility usage data for the extension equipment is not yet available. A reactor volume scale-up factor of 3.54x has therefore been applied throughout as a conservative feasibility-level estimate. • This approach will be refined as process design progresses and usage data becomes available at later design stages.		References	
Summary See specific calculation sheets			
Assumptions See specific calculation sheets			

Revision: A1



Line No.	Calculation															Notes
1	Scale-up for reactor volume and utility usage assessment															Assumption using volume scale-up factor is based on unavailability of utility usage data for equipment in new extension facility
2																
3	1) To estimate the scale-up in reactor volumes between the existing facility and new extension facility															
4	Scenario: In order to estimate the utility usage in the new extension facility (as this is not known), a scale-up factor will be applied for the process utilities. This will be based on the ratio between the total reactor volumes of the existing facility and the new extension facility. Utility usages for the existing facility will be multiplied by this scale-up factor to estimate the utility usage by the new extension facility.															
5																
6																
7	Existing Facility															Existing facility equipment data obtained from Pilot Plant Equipment List 15 March 2026
8		Tag No.	Facility Room No.	Volume (L)												
9	Reactors	R01	P2.11	250												
10		R02	P2.11	250												
11		R03	P2.11	630												
12		R04	P2.11	630												
13		R05	P2.11	1600												
14		R06	P2.11	1600												
15		R07	P2.09	250												
16		R08	P2.09	630												
17		R09	P2.09	630												
18		R10	P2.09	1600												
19		R11	P2.09	1600												
20		R12	P2.09	2500												
21		R13	P2.03	630												
22		R14	P2.03	250												
23		R15	P2.03	1600												
24		R16	P2.03	630												
25		R17	P2.03	2500												
26		R18	P2.03	1600												
27		R19	P2.05	630												
28		R20	P2.05	250												
29		R21	P2.05	1600												
30		R22	P2.05	630												
31	Hydrogenation	R23	P2.05	250												
32	Filter Dryers	MFD01	P1.13	125												
33		MFD02	P1.13	300												
34		FD02-RC1	PG.05	250												
35		FD03-RC1	PG.09	750												
36		FD04-RC1	PG.09	300												
37	FD05-RC1	PG.03	600													
38	Total			25065 L												
39				Total volume of vessels in existing facility												
40																
41	New Extension Facility															New extension facility equipment data obtained from SW-902Expansion-Equipment_List-20 March
42		Tag No.	Facility Room No.	Volume (L)												
43	Reactors	R25	TBC	5000												
44		R26	TBC	8000												
45		R27	TBC	5000												
46		R28	TBC	8000												
47		R29	TBC	2500												
48		R30	TBC	2500												
49		R31	TBC	5000												
50		R32	TBC	5000												
51		R33	TBC	8000												
52		R34	TBC	8000												
53	Hydrogenation	R35	TBC	2500												
54	Filter Dryers	FD01	TBC	1130												
55		FD02	TBC	1530												
56		FD03	TBC	1530												
57	Total			63690 L												
58				Total volume of vessels in new extension facility												
59																
60																
61	Reactor Volume Scale-up Ratio															
62	Existing facility total reactor volume			25065 L												
63	New extension facility total reactor volume			63690 L												
64	Reactor scale-up ratio			3.54												
65																

Project:Asymchem PDF Extension Facility Project No:300291 Subject:Utility Capacity Assessment Calculations By:ME Approved:ME Checked:MC Date:21-May-26 Document No:300291-PR-CA-0001 Sheet Name:Hot HTF - Process Revision:A1																
Line No.	Calculation													Notes		
1	Heating Loads													Information on material properties obtained from engineeringtoolbox		
2																
3	HTF (Heat Transfer Fluid) is the cold/hot utility used. In this case, HTF fluid is Syltherm XLT.															
4	1) To estimate the maximum delta T that can be achieved for hot HTF utility based on the existing heater															
5	Scenario: The existing pilot plant has a 510kW heater which heats HTF from the HTF tank to obtain 130C hot HTF for maintaining the temperature of the facility reactor vessels. The existing hot HTF supply pump has a fixed pumping rate of 50m3/hr which supplies hot HTF to the reactor jackets.															
6																
7	Existing Hot HTF heater heat capacity		510	kW												
8	Hot HTF heat capacity		2.024	kJ/kg°C	assumption											
9	Hot HTF density		732	g/cm³	assumption											
10	Flowrate		50	m³/hr												
11			10.17	kg/s												
12	Supply Temperature		130	°C												
13																
14	Maximum Delta T Hot HTF		24.78	°C												
15																
16	Minimum return T Hot HTF		105.2	°C												
17																
18																
19	2) To calculate the hot HTF flowrate required to achieve desired temperature control in the solvent reactors															
20																
21	Scenario: Hot HTF is used to control temperature of solvent mixture in the reactor to a specific range of 0.5C - 1C/min. Hot HTF is pumped to the reactor jackets to achieve this temperature control where it is supplied at 130C and returns at 105C. Solvents used are acetone and methanol															
22																
23	Temperature control to achieve in reactors		0.5	C/min												
24			1.0	C/min												
25	Existing Hot HTF heater heat capacity		510	kW												
26	Hot HTF heat capacity		2.024	kJ/kg°C	assumption											
27	Solvent:	Acetone heat capacity	2.14	kJ/kg°C												
28		Methanol heat capacity	2.8	kJ/kg°C												
29	Hot HTF density		732	g/cm³	assumption											
30	Solvent:	Acetone density	785	g/cm³												
31		Methanol density	753	g/cm³												
32	Hot HTF Supply Temperature		130	°C												
33	Hot HTF Return Temperature		105.2	°C												
34																
35	Scenario is based on 2500L reactor. This is the largest reactor volume in the existing facility so is considered the worst-case/ largest heating load here.															
36	For this calculation, the worst-case scenario will be based on the largest value for density and heat capacity for the solvents. Methanol has higher heat capacity (2.8 kJ/kg°C) and acetone has higher density at 785 kg/m³)															
37																
38	Reactor volume		2500	L	Vessel filling volume for process		100%									
39	Mass of solvent in reactor		1963	kg												
40																
41	0.5°C/min reactor temperature control															
42	Hot HTF cooling load		45.8	kW												
43	Hot HTF flowrate		0.91	kg/s			4.5	m³/hr								
44																
45	1°C/min reactor temperature control															
46	Hot HTF cooling load		91.6	kW												
47	Hot HTF flowrate		1.83	kg/s			9.0	m³/hr								
48																
49																
50	3) Hot HTF diversity assessment on all reactors in existing and new extension facility															
51																
52	An assessment on the hot HTF diversity is required to determine the operability of the new extension facility with regards to hot HTF used for reactor temperature control. This will be based on the existing facility reactor volumes. A scale-up will be applied for the new extension facility based on the reactor volumes to estimate potential diversity in hot HTF usage.															
53																
54	0.5°C/min reactor temperature control															
55																
56	Existing Facility Reactors				New Extension Facility Reactors											
57	Tag No.	Reactor Volume (L)	Heat Load (kW)	Tag No.	Reactor Volume (L)	Heat Load (kW)										
58	R01	250	4.6	R25	5000	91.6										

59		R02	250	4.6		R26	8000	146.5		
60		R03	630	11.5		R27	5000	91.6		
61		R04	630	11.5		R28	8000	146.5		
62		R05	1600	29.3		R29	2500	45.8		
63		R06	1600	29.3		R30	2500	45.8		
64		R07	250	4.6		R31	5000	91.6		
65		R08	630	11.5		R32	5000	91.6		
66		R09	630	11.5		R33	8000	146.5		

67		R10	1600	29.3		R34	8000	146.5		
68		R11	1600	29.3		R35	2500	45.8		
69		R12	2500	45.8		FD01	1130	20.7		
70		R13	630	11.5		FD02	1530	28.0		
71		R14	250	4.6		FD03	1530	28.0		
72		R15	1600	29.3			Total	1075	kW	
73		R16	630	11.5						
74		R17	2500	45.8						
75		R18	1600	29.3						
76		R19	630	11.5						
77		R20	250	4.6						
78		R21	1600	29.3						
79		R22	630	11.5						
80		R23	250	4.6						
81		MFD01	125	2.3						
82		MFD02	300	5.5						
83		FD02-RC1	250	4.6						
84		FD03-RC1	750	13.7						
85		FD04-RC1	300	5.5						
86		FD05-RC1	600	11.0						
87		Total	459	kW						

1°C/min reactor temperature control

91	Existing Facility Reactors			New Extension Facility Reactors		
92	Tag No.	Reactor Volume (L)	Heat Load (kW)	Tag No.	Reactor Volume (L)	Heat Load (kW)
93	R01	250	9.2	R25	5000	183.2
94	R02	250	9.2	R26	8000	293.1
95	R03	630	23.1	R27	5000	183.2
96	R04	630	23.1	R28	8000	293.1
97	R05	1600	58.6	R29	2500	91.6
98	R06	1600	58.6	R30	2500	91.6
99	R07	250	9.2	R31	5000	183.2
100	R08	630	23.1	R32	5000	183.2
101	R09	630	23.1	R33	8000	293.1
102	R10	1600	58.6	R34	8000	293.1
103	R11	1600	58.6	R35	2500	91.6
104	R12	2500	91.6	FD01	1130	41.4
105	R13	630	23.1	FD02	1530	56.0
106	R14	250	9.2	FD03	1530	56.0
107	R15	1600	58.6	Total	2150	kW
108	R16	630	23.1			
109	R17	2500	91.6			
110	R18	1600	58.6			
111	R19	630	23.1			
112	R20	250	9.2			
113	R21	1600	58.6			
114	R22	630	23.1			
115	R23	250	9.2			
116	MFD01	125	4.6			
117	MFD02	300	11.0			
118	FD02-RC1	250	9.2			
119	FD03-RC1	750	27.5			
120	FD04-RC1	300	11.0			
121	FD05-RC1	600	22.0			
122	Total	918	kW			

125	Tag	Existing or New Extension?	Reactor Volume	Heat Load @ 0.5°C/min	Heat Load @ 1°C/min	Is Reactor Active?	Desired Temp Control	Active Heat Load (kW)
126	R01	Existing	250	4.6	9.2	☐	1°C/min	0.0
127	R02	Existing	250	4.6	9.2	☐	1°C/min	0.0
128	R03	Existing	630	11.5	23.1	☐	1°C/min	0.0
129	R04	Existing	630	11.5	23.1	☐	1°C/min	0.0
130	R05	Existing	1600	29.3	58.6	☐	1°C/min	0.0
131	R06	Existing	1600	29.3	58.6	☐	1°C/min	0.0
132	R07	Existing	250	4.6	9.2	☐	1°C/min	0.0
133	R08	Existing	630	11.5	23.1	☐	1°C/min	0.0
134	R09	Existing	630	11.5	23.1	☐	1°C/min	0.0
135	R10	Existing	1600	29.3	58.6	☒	1°C/min	58.6
136	R11	Existing	1600	29.3	58.6	☒	1°C/min	58.6
137	R12	Existing	2500	45.8	91.6	☒	1°C/min	91.6
138	R13	Existing	630	11.5	23.1	☐	1°C/min	0.0
139	R14	Existing	250	4.6	9.2	☐	1°C/min	0.0
140	R15	Existing	1600	29.3	58.6	☐	1°C/min	0.0
141	R16	Existing	630	11.5	23.1	☐	1°C/min	0.0
142	R17	Existing	2500	45.8	91.6	☒	1°C/min	91.6

Typical reactor operation is:
Existing - 6 equipment trains at same time
Design based on 6 largest reactors in use

New extension - 3 equipment trains at same time
Design based on 2x 5000L and 1x 8000L
reactors in use

143		R18	Existing	1600	29.3	58.6	☒	1°C/min	58.6	
144		R19	Existing	630	11.5	23.1	☐	1°C/min	0.0	
145		R20	Existing	250	4.6	9.2	☐	1°C/min	0.0	
146		R21	Existing	1600	29.3	58.6	☒	1°C/min	58.6	
147		R22	Existing	630	11.5	23.1	☒	1°C/min	23.1	
148		R23	Existing	250	4.6	9.2	☐	1°C/min	0.0	
149		MFD01	Existing	125	2.3	4.6	☐	1°C/min	0.0	
150		MFD02	Existing	300	5.5	11.0	☒	1°C/min	11.0	
151		FD02-RC1	Existing	250	4.6	9.2	☐	1°C/min	0.0	
152		FD03-RC1	Existing	750	13.7	27.5	☐	1°C/min	0.0	
153		FD04-RC1	Existing	300	5.5	11.0	☐	1°C/min	0.0	
154		FD05-RC1	Existing	600	11.0	22.0	☒	1°C/min	22.0	
155		R25	New Extension	5000	91.6	183.2	☒	1°C/min	183.2	
156		R26	New Extension	8000	146.5	293.1	☒	1°C/min	293.1	
157		R27	New Extension	5000	91.6	183.2	☒	1°C/min	183.2	
158		R28	New Extension	8000	146.5	293.1	☐	1°C/min	0.0	
159		R29	New Extension	2500	45.8	91.6	☐	1°C/min	0.0	
160		R30	New Extension	2500	45.8	91.6	☐	1°C/min	0.0	
161		R31	New Extension	5000	91.6	183.2	☐	1°C/min	0.0	
162		R32	New Extension	5000	91.6	183.2	☐	1°C/min	0.0	
163		R33	New Extension	8000	146.5	293.1	☐	1°C/min	0.0	
164		R34	New Extension	8000	146.5	293.1	☐	1°C/min	0.0	
165		R35	New Extension	2500	45.8	91.6	☐	1°C/min	0.0	
166		FD01	New Extension	1130	20.7	41.4	☐	1°C/min	0.0	
167		FD02	New Extension	1530	28.0	56.0	☒	1°C/min	56.0	
168		FD03	New Extension	1530	28.0	56.0	☐	1°C/min	0.0	
169										
170										
171										
172		Total Simultaneous Active Heat Load			1189	kW	Recommendation: Separate 900 kW heater dedicated for extension alone required. Additionally, new recirculation loop for hot HTF with dedicated mag drive pumps for extension facility			
173		Maximum Heater Capacity			510	kW				
174		Spare Capacity			-679	kW				
175		Capacity Status			EXCEEDS HEATER CAPACITY					
176										
177										
178		Summary								
179		Existing reactors active load			474	kW				
180		New extension reactors active load			715	kW				
181		Total simultaneous load			1189	kW				
182		Heater capacity			510	kW				
183		Spare capacity			-679	kW				
184		% utilisation			233	%				
185										

Project:Asymchem PDF Extension Facility Project No:300291 Subject:Utility Capacity Assessment Calculations By:MEChecked:MC Approved:MEDate:21-May-26 Document No:300291-PR-CA-0001 Sheet Name:Hot HTF - Cleaning Revision:A1																	
Line No.	Calculation															Notes	
1	Heating Loads															Information on material properties obtained from engineeringtoolbox	
2																	
3	HTF (Heat Transfer Fluid) is the cold/hot utility used. In this case, HTF fluid is Syltherm XLT.																
4	1) To estimate the maximum delta T that can be achieved for hot HTF utility based on the existing heater																
5	Scenario: The existing pilot plant has a 510kW heater which heats HTF from the HTF tank to obtain 130C hot HTF for maintaining the temperature of the facility reactor vessels. The existing hot HTF supply pump has a fixed pumping rate of 50m3/hr which supplies hot HTF to the reactor jackets.																
6																	
7	Existing Hot HTF heater heat capacity		510	kW													
8	Hot HTF heat capacity		2.024	kJ/kg°C	assumption												
9	Hot HTF density		732	g/cm³	assumption												
10	Flowrate		50	m³/hr													
11			10.17	kg/s													
12	Supply Temperature		130	°C													
13																	
14	Maximum Delta T Hot HTF		24.78	°C													
15																	
16	Minimum return T Hot HTF		105.2	°C													
17																	
18																	
19																	
20	2) To calculate the hot HTF flowrate required to achieve desired temperature control in the solvent reactors																
21	Scenario: Hot HTF is used to control temperature of solvent mixture in the reactor to a specific range of 0.5C - 1C/min. Hot HTF is pumped to the reactor jackets to achieve this temperature control where it is supplied at 130C and returns at 105C. Solvents used are acetone and methanol																
22																	
23	Temperature control to achieve in reactors		0.5	C/min													
24			1.0	C/min													
25	Existing Hot HTF heater heat capacity		510	kW													
26	Hot HTF heat capacity		2.024	kJ/kg°C	assumption												
27	Solvent: Acetone heat capacity		2.14	kJ/kg°C													
28	Methanol heat capacity		2.8	kJ/kg°C													
29	Hot HTF density		732	g/cm³	assumption												
30	Solvent: Acetone density		785	g/cm³													
31	Methanol density		753	g/cm³													
32	Hot HTF Supply Temperature		130	°C													
33	Hot HTF Return Temperature		105.2	°C													
34																	
35	Scenario is based on 2500L reactor. This is the largest reactor volume in the existing facility so is considered the worst-case/ largest heating load here.																
36	For this calculation, the worst-case scenario will be based on the largest value for density and heat capacity for the solvents. Methanol has higher heat capacity (2.8 kJ/kg°C) and acetone has higher density at 785 kg/m³)																
37																	
38	Reactor volume		2500	L	Vessel filling volume for process		25%										
39	Mass of solvent in reactor		491	kg													
40																	
41	0.5°C/min reactor temperature control																
42	Hot HTF cooling load		11.4	kW													
43	Hot HTF flowrate		0.23	kg/s			1.1	m³/hr									
44																	
45	1°C/min reactor temperature control																
46	Hot HTF cooling load		22.9	kW													
47	Hot HTF flowrate		0.46	kg/s			2.2	m³/hr									
48																	
49																	
50																	
51	3) Hot HTF diversity assessment on all reactors in existing and new extension facility																
52	An assessment on the hot HTF diversity is required to determine the operability of the new extension facility with regards to hot HTF used for reactor temperature control. This will be based on the existing facility reactor volumes. A scale-up will be applied for the new extension facility based on the reactor volumes to estimate potential diversity in hot HTF usage.																
53																	
54	0.5°C/min reactor temperature control																
55																	
56	Existing Facility Reactors					New Extension Facility Reactors											
57	Tag No.	Reactor Volume (L)	Heat Load (kW)			Tag No.	Reactor Volume (L)	Heat Load (kW)									

58		R01	250	1.1		R25	5000	22.9		
59		R02	250	1.1		R26	8000	36.6		
60		R03	630	2.9		R27	5000	22.9		
61		R04	630	2.9		R28	8000	36.6		
62		R05	1600	7.3		R29	2500	11.4		
63		R06	1600	7.3		R30	2500	11.4		
64		R07	250	1.1		R31	5000	22.9		
65		R08	630	2.9		R32	5000	22.9		
66		R09	630	2.9		R34	8000	36.6		

67	R10	1600	7.3		R35	8000	36.6	
68	R11	1600	7.3			Total	261	kW
69	R12	2500	11.4					
70	R13	630	2.9					
71	R14	250	1.1					
72	R15	1600	7.3					
73	R16	630	2.9					
74	R17	2500	11.4					
75	R18	1600	7.3					
76	R19	630	2.9					
77	R20	250	1.1					
78	R21	1600	7.3					
79	R22	630	2.9					
80		Total	103	kW				

1°C/min reactor temperature control

	Existing Facility Reactors			New Extension Facility Reactors		
85	Tag No.	Reactor Volume (L)	Heat Load (kW)	Tag No.	Reactor Volume (L)	Heat Load (kW)
86	R01	250	2.3	R25	5000	45.8
87	R02	250	2.3	R26	8000	73.3
88	R03	630	5.8	R27	5000	45.8
89	R04	630	5.8	R28	8000	73.3
90	R05	1600	14.7	R29	2500	22.9
91	R06	1600	14.7	R30	2500	22.9
92	R07	250	2.3	R31	5000	45.8
93	R08	630	5.8	R32	5000	45.8
94	R09	630	5.8	R34	8000	73.3
95	R10	1600	14.7	R35	8000	73.3
96	R11	1600	14.7		Total	522 kW
97	R12	2500	22.9			
98	R13	630	5.8			
99	R14	250	2.3			
100	R15	1600	14.7			
101	R16	630	5.8			
102	R17	2500	22.9			
103	R18	1600	14.7			
104	R19	630	5.8			
105	R20	250	2.3			
106	R21	1600	14.7			
107	R22	630	5.8			
108		Total	206 kW			

110	Tag	Existing or New Extension?	Reactor Volume	Heat Load @ 0.5°C/min	Heat Load @ 1°C/min	Is Reactor Active?	Desired Temp Control	Active Heat Load (kW)
111	R01	Existing	250	1.1	2.3	☐	1°C/min	0.0
112	R02	Existing	250	1.1	2.3	☐	1°C/min	0.0
113	R03	Existing	630	2.9	5.8	☐	1°C/min	0.0
114	R04	Existing	630	2.9	5.8	☐	1°C/min	0.0
115	R05	Existing	1600	7.3	14.7	☒	1°C/min	14.7
116	R06	Existing	1600	7.3	14.7	☒	1°C/min	14.7
117	R07	Existing	250	1.1	2.3	☐	1°C/min	0.0
118	R08	Existing	630	2.9	5.8	☐	1°C/min	0.0
119	R09	Existing	630	2.9	5.8	☐	1°C/min	0.0
120	R10	Existing	1600	7.3	14.7	☒	1°C/min	14.7
121	R11	Existing	1600	7.3	14.7	☒	1°C/min	14.7
122	R12	Existing	2500	11.4	22.9	☒	1°C/min	22.9
123	R13	Existing	630	2.9	5.8	☐	1°C/min	0.0
124	R14	Existing	250	1.1	2.3	☐	1°C/min	0.0
125	R15	Existing	1600	7.3	14.7	☐	1°C/min	0.0
126	R16	Existing	630	2.9	5.8	☐	1°C/min	0.0
127	R17	Existing	2500	11.4	22.9	☒	1°C/min	22.9
128	R18	Existing	1600	7.3	14.7	☐	1°C/min	0.0
129	R19	Existing	630	2.9	5.8	☐	1°C/min	0.0
130	R20	Existing	250	1.1	2.3	☐	1°C/min	0.0
131	R21	Existing	1600	7.3	14.7	☐	1°C/min	0.0
132	R22	Existing	630	2.9	5.8	☐	1°C/min	0.0
133	R25	New Extension	5000	22.9	45.8	☒	1°C/min	45.8
134	R26	New Extension	8000	36.6	73.3	☒	1°C/min	73.3
135	R27	New Extension	5000	22.9	45.8	☒	1°C/min	45.8
136	R28	New Extension	8000	36.6	73.3	☐	1°C/min	0.0
137	R29	New Extension	2500	11.4	22.9	☐	1°C/min	0.0
138	R30	New Extension	2500	11.4	22.9	☐	1°C/min	0.0
139	R31	New Extension	5000	22.9	45.8	☐	1°C/min	0.0
140	R32	New Extension	5000	22.9	45.8	☐	1°C/min	0.0
141	R33	New Extension	8000	36.6	73.3	☐	1°C/min	0.0
142	R34	New Extension	8000	36.6	73.3	☐	1°C/min	0.0

145	Total Simultaneous Active Heat Load	269	kW
146	Maximum Heater Capacity	510	kW
147	Spare Capacity	241	kW
148	Capacity Status	WITHIN CAPACITY	

Typical reactor operation is:
Existing - 6 equipment trains at same time
Design based on 6 largest reactors in use

New extension - 3 equipment trains at same time
Design based on 2x 5000L and 1x 8000L
reactors in use

149									
150			Summary						
151			Existing reactors active load	104	kW				
152			New extension reactors active load	165	kW				
153			Total simultaneous load	269	kW				
154			Heater capacity	510	kW				
155			Spare capacity	241	kW				
156			% utilisation	53	%				
157									

Project:Asymchem PDF Extension Facility Project No:300291 Subject:Utility Capacity Assessment Calculations By:ME Approved:ME Checked:MC Date:21-May-26 Document No:300291-PR-CA-0001 Sheet Name:Cold HTF - Process Revision:A1																								
Line No.	Calculation															Notes								
1	Heating Loads															Information on material properties obtained from engineeringtoolbox								
2																								
3	HTF (Heat Transfer Fluid) is the cold/hot utility used. In this case, HTF fluid is Syltherm XLT.																							
4	1) To calculate the Chiller load for cold HTF for the existing facility and estimate spare capacity available for the new expansion																							
5	Scenario: The existing pilot plant has a 780kW ammonia chiller which cools cold HTF from -20C to -25C for maintaining the temperature of the facility reactor vessels. The existing cold HTF supply pump has a fixed pumping rate of 100m3/hr which supplies cold HTF to the reactor jackets.																							
6																								
7	Existing cold HTF chiller heat capacity		780	kW																				
8	Cold HTF heat capacity		1.7	kJ/kg°C																				
9	Cold HTF density		896.4	g/cm³																				
10	Flowrate		100	m³/hr																				
11	Supply Temperature		-18	°C		Into chiller																		
12	Return Temperature		-25	°C		From chiller to reactor jackets																		
13	Temperature rise		7	°C																				
14																								
15	Chiller heat load			294.2	kW																			
16																								
17																								
18	2) To calculate the cold HTF flowrate required to achieve desired temperature control in the solvent reactors																							
19																								
20	Scenario: Cold HTF is used to control temperature of solvent mixture in the reactor to a specific range of 0.5C - 1C/min. This is due to the exothermic nature of the solvent reactors. Cold HTF is pumped to the reactor jackets to achieve this temperature control where it is supplied at -25C and returns at -20C. Solvents used are acetone and methanol																							
21																								
22	Temperature control to achieve in reactors		0.5	C/min																				
23			1.0	C/min																				
24	Existing cold HTF chiller heat capacity		780	kW																				
25	Cold HTF heat capacity		1.7	kJ/kg°C																				
26	Solvent:	Acetone heat capacity	2.14	kJ/kg°C																				
27		Methanol heat capacity	2.8	kJ/kg°C																				
28	Cold HTF density		896.4	g/cm³																				
29	Solvent:	Acetone density	785	g/cm³																				
30		Methanol density	753	g/cm³																				
31	Cold HTF Supply Temperature		-25	°C																				
32	Cold HTF Return Temperature		-20	°C																				
33																								
34	Scenario is based on 2500L reactor. This is the largest reactor volume in the existing facility so is considered the worst-case/ largest cooling load here.																							
35	For this calculation, the worst-case scenario will be based on the largest value for density and heat capacity for the solvents. Methanol has higher heat capacity (2.8 kJ/kg°C) and acetone has higher density at 785 kg/m³)																							
36																								
37	Reactor volume		2500	L	Vessel filling volume for process		100%																	
38	Mass of solvent in reactor		1963	kg																				
39																								
40	0.5°C/min reactor temperature control																							
41	Cold HTF cooling load			45.8	kW																			
42	Cold HTF flowrate			5.43	kg/s			21.8	m³/hr															
43																								
44	1°C/min reactor temperature control																							
45	Cold HTF cooling load			91.6	kW																			
46	Cold HTF flowrate			10.85	kg/s			43.6	m³/hr															
47																								
48																								
49	3) To calculate the maximum cold HTF cooling for the chiller based on the worst-case 8000L reactor crash cooling																							
50																								
51	Currently, crash cooling load is undetermined due to no knowledge of reaction exotherms at this stage. When crash cooling is required, the inlet jacket valve is fully opened and the reaction is flooded with cold HTF until it is sufficiently cooled. It is assumed that there is sufficient cold HTF to maintain this crash cooling load for the new extension reactors. This will be assessed further at the next stage.																							
52																								
53																								
54																								
55																								
56																								
57	4) Cold HTF diversity assessment on all reactors in existing and new extension facility																							
58																								

An assessment on the cold HTF diversity is required to determine the operability of the new extension facility with regards to cold HTF used for reactor temperature control. This will be based on the existing facility reactor volumes. A scale-up will be applied for the new extension facility based on the reactor volumes to estimate potential diversity in cold HTF usage

0.5°C/min reactor temperature control

Existing Facility Reactors			New Extension Facility Reactors		
Tag No.	Reactor Volume (L)	Heat Load (kW)	Tag No.	Reactor Volume (L)	Heat Load (kW)
R01	250	4.6	R25	5000	91.6
R02	250	4.6	R26	8000	146.5
R03	630	11.5	R27	5000	91.6
R04	630	11.5	R28	8000	146.5
R05	1600	29.3	R29	2500	45.8
R06	1600	29.3	R30	2500	45.8
R07	250	4.6	R31	5000	91.6
R08	630	11.5	R32	5000	91.6
R09	630	11.5	R33	8000	146.5
R10	1600	29.3	R34	8000	146.5
R11	1600	29.3	R35	2500	45.8
R12	2500	45.8	FD01	1130	20.7
R13	630	11.5	FD02	1530	28.0
R14	250	4.6	FD03	1530	28.0
R15	1600	29.3	Total		1075 kW
R16	630	11.5			
R17	2500	45.8			
R18	1600	29.3			
R19	630	11.5			
R20	250	4.6			
R21	1600	29.3			
R22	630	11.5			
R23	250	4.6			
MFD01	125	2.3			
MFD02	300	5.5			
FD02-RC1	250	4.6			
FD03-RC1	750	13.7			
FD04-RC1	300	5.5			
FD05-RC1	600	11.0			
Total		412 kW			

1°C/min reactor temperature control

Existing Facility Reactors			New Extension Facility Reactors		
Tag No.	Reactor Volume (L)	Heat Load (kW)	Tag No.	Reactor Volume (L)	Heat Load (kW)
R01	250	9.2	R25	5000	183.2
R02	250	9.2	R26	8000	293.1
R03	630	23.1	R27	5000	183.2
R04	630	23.1	R28	8000	293.1
R05	1600	58.6	R29	2500	91.6
R06	1600	58.6	R30	2500	91.6
R07	250	9.2	R31	5000	183.2
R08	630	23.1	R32	5000	183.2
R09	630	23.1	R33	8000	293.1
R10	1600	58.6	R34	8000	293.1
R11	1600	58.6	R35	2500	91.6
R12	2500	91.6	FD01	1130	41.4
R13	630	23.1	FD02	1530	56.0
R14	250	9.2	FD03	1530	56.0
R15	1600	58.6	Total		2150 kW
R16	630	23.1			
R17	2500	91.6			
R18	1600	58.6			
R19	630	23.1			
R20	250	9.2			
R21	1600	58.6			
R22	630	23.1			
R23	250	9.2			
MFD01	125	4.6			
MFD02	300	11.0			
FD02-RC1	250	9.2			
FD03-RC1	750	27.5			
FD04-RC1	300	11.0			
FD05-RC1	600	22.0			
Total		824 kW			

Tag	Existing or New Extension?	Reactor Volume	Heat Load @ 0.5°C/min	Heat Load @ 1°C/min	Is Reactor Active?	Desired Temp Control	Active Heat Load (kW)
R01	Existing	250	4.6	9.2	☐	0.5°C/min	0.0
R02	Existing	250	4.6	9.2	☐	0.5°C/min	0.0
R03	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R04	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R05	Existing	1600	29.3	58.6	☐	0.5°C/min	0.0
R06	Existing	1600	29.3	58.6	☐	0.5°C/min	0.0

Typical reactor operation is:
Existing - 6 equipment trains at same time
Design based on 6 largest reactors in use

New extension - 3 equipment trains at same time
Design based on 2x 5000L and 1x 8000L reactors in use

139		R07	Existing	250	4.6	9.2	☐	0.5°C/min	0.0	
140		R08	Existing	630	11.5	23.1	☐	0.5°C/min	0.0	
141		R09	Existing	630	11.5	23.1	☐	0.5°C/min	0.0	
142		R10	Existing	1600	29.3	58.6	☒	0.5°C/min	29.3	
143		R11	Existing	1600	29.3	58.6	☒	0.5°C/min	29.3	
144		R12	Existing	2500	45.8	91.6	☒	0.5°C/min	45.8	
145		R13	Existing	630	11.5	23.1	☐	0.5°C/min	0.0	
146		R14	Existing	250	4.6	9.2	☐	0.5°C/min	0.0	
147		R15	Existing	1600	29.3	58.6	☒	0.5°C/min	29.3	
148		R16	Existing	630	11.5	23.1	☐	0.5°C/min	0.0	
149		R17	Existing	2500	45.8	91.6	☐	0.5°C/min	0.0	
150		R18	Existing	1600	29.3	58.6	☐	0.5°C/min	0.0	
151		R19	Existing	630	11.5	23.1	☐	0.5°C/min	0.0	
152		R20	Existing	250	4.6	9.2	☐	0.5°C/min	0.0	
153		R21	Existing	1600	29.3	58.6	☐	0.5°C/min	0.0	
154		R22	Existing	630	11.5	23.1	☐	0.5°C/min	0.0	
155		R23	Existing	250	4.6	9.2	☐	1°C/min	0.0	
156		MFD01	Existing	125	2.3	4.6	☐	1°C/min	0.0	
157		MFD02	Existing	300	5.5	11.0	☐	1°C/min	0.0	
158		FD02-RC1	Existing	250	4.6	9.2	☐	1°C/min	0.0	
159		FD03-RC1	Existing	750	13.7	27.5	☐	1°C/min	0.0	
160		FD04-RC1	Existing	300	5.5	11.0	☐	1°C/min	0.0	
161		FD05-RC1	Existing	600	11.0	22.0	☐	1°C/min	0.0	
162		R25	New Extension	5000	91.6	183.2	☒	0.5°C/min	91.6	
163		R26	New Extension	8000	146.5	293.1	☒	0.5°C/min	146.5	
164		R27	New Extension	5000	91.6	183.2	☒	0.5°C/min	91.6	
165		R28	New Extension	8000	146.5	293.1	☐	0.5°C/min	0.0	
166		R29	New Extension	2500	45.8	91.6	☐	0.5°C/min	0.0	
167		R30	New Extension	2500	45.8	91.6	☐	0.5°C/min	0.0	
168		R31	New Extension	5000	91.6	183.2	☐	0.5°C/min	0.0	
169		R32	New Extension	5000	91.6	183.2	☐	0.5°C/min	0.0	
170		R33	New Extension	8000	146.5	293.1	☐	0.5°C/min	0.0	
171		R34	New Extension	8000	146.5	293.1	☐	0.5°C/min	0.0	
172		R35	New Extension	2500	45.8	91.6	☐	1°C/min	0.0	
173		FD01	New Extension	1130	20.7	41.4	☐	1°C/min	0.0	
174		FD02	New Extension	1530	28.0	56.0	☒	1°C/min	56.0	
175		FD03	New Extension	1530	28.0	56.0	☐	1°C/min	0.0	
176										
177										
178										
179		Total Simultaneous Active Heat Load		519	kW					
180		Maximum Chiller Capacity		780	kW					
181		Spare Capacity		261	kW					
182		Capacity Status		WITHIN CAPACITY						
183										
184										
185		Summary								
186		Existing reactors active load		134	kW					
187		New extension reactors active load		386	kW					
188		Total simultaneous load		519	kW					
189		Maximum chiller capacity		780	kW					
190		Spare capacity		261	kW					
191		% utilisation		67	%					
192										

Project:Asymchem PDF Extension Facility Project No:300291 Subject:Utility Capacity Assessment Calculations By:ME Approved:ME Checked:MC Date:21-May-26 Document No:300291-PR-CA-0001 Sheet Name:Cold HTF - Cleaning Revision:A1																								
Line No.	Calculation															Notes								
1	Heating Loads															Information on material properties obtained from engineeringtoolbox								
2																								
3	HTF (Heat Transfer Fluid) is the cold/hot utility used. In this case, HTF fluid is Syltherm XLT.																							
4	1) To calculate the Chiller load for cold HTF for the existing facility and estimate spare capacity available for the new expansion															Cold HTF heat capacity at -20C Cold HTF density at -20C Based on existing supply pump capacity								
5	Scenario: The existing pilot plant has a 780kW ammonia chiller which cools cold HTF from -20C to -25C for maintaining the temperature of the facility reactor vessels. The existing cold HTF supply pump has a fixed pumping rate of 100m3/hr which supplies cold HTF to the reactor jackets.																							
6																								
7	Existing cold HTF chiller heat capacity		780	kW																				
8	Cold HTF heat capacity		1.7	kJ/kg°C																				
9	Cold HTF density		896.4	g/cm³																				
10	Flowrate		100	m³/hr																				
11	Supply Temperature		-18	°C		Into chiller																		
12	Return Temperature		-25	°C		From chiller to reactor jackets																		
13	Temperature rise		7	°C																				
14																								
15	Chiller heat load			294.2	kW																			
16																								
17																								
18	2) To calculate the cold HTF flowrate required to achieve desired temperature control in the solvent reactors															Cold HTF heat capacity at -20C Acetone heat capacity at standard conditions Methanol heat capacity at standard conditions Cold HTF density at -20C Acetone density at standard conditions Methanol density at standard conditions								
19																								
20	Scenario: Cold HTF is used to control temperature of solvent mixture in the reactor to a specific range of 0.5C - 1C/min. This is due to the exothermic nature of the solvent reactors. Cold HTF is pumped to the reactor jackets to achieve this temperature control where it is supplied at -25C and returns at -20C. Solvents used are acetone and methanol																							
21																								
22	Temperature control to achieve in reactors		0.5	C/min																				
23			1.0	C/min																				
24	Existing cold HTF chiller heat capacity		0	kW																				
25	Cold HTF heat capacity		1.7	kJ/kg°C																				
26	Solvent: Acetone heat capacity		2.14	kJ/kg°C																				
27	Methanol heat capacity		2.8	kJ/kg°C																				
28	Cold HTF density		896.4	g/cm³																				
29	Solvent: Acetone density		785	g/cm³																				
30	Methanol density		753	g/cm³																				
31	Cold HTF Supply Temperature		-25	°C																				
32	Cold HTF Return Temperature		-20	°C																				
33																								
34	Scenario is based on 2500L reactor. This is the largest reactor volume in the existing facility so is considered the worst-case/ largest cooling load here.															Vessel 1/2 full for solvent cleaning Based on acetone density 0.5°C/min reactor temperature 1°C/min reactor temperature								
35	For this calculation, the worst-case scenario will be based on the largest value for density and heat capacity for the solvents. Methanol has higher heat capacity (2.8 kJ/kg°C) and acetone has higher density at 785 kg/m³)																							
36																								
37	Reactor volume		2500	L		Vessel filling volume for process		50%																
38	Mass of solvent in reactor		981	kg																				
39																								
40	0.5°C/min reactor temperature control																							
41	Cold HTF cooling load			22.9	kW																			
42	Cold HTF flowrate			2.71	kg/s			10.9	m³/hr															
43																								
44	1°C/min reactor temperature control																							
45	Cold HTF cooling load			45.8	kW																			
46	Cold HTF flowrate			5.43	kg/s			21.8	m³/hr															
47																								
48																								
	3) To calculate the maximum cold HTF cooling for the chiller based on the worst-case 8000L reactor crash cooling															Cold HTF heat capacity at -20C Cold HTF density at -20C Based on supply pump capacity								
	Existing cold HTF chiller heat capacity			kW																				
	Cold HTF heat capacity		1.7	kJ/kg°C																				
	Cold HTF density		896.4	g/cm³																				
	Flowrate		100	m³/hr																				
	Supply Temperature		-20	°C																				
	Return Temperature		-25	°C																				
	Temperature rise		5	°C																				

125		R15	Existing	1600	14.7	29.3	☒	1°C/min	29.3	
126		R16	Existing	630	5.8	11.5	☐	1°C/min	0.0	
127		R17	Existing	2500	22.9	45.8	☒	1°C/min	45.8	
128		R18	Existing	1600	14.7	29.3	☒	1°C/min	29.3	
129		R19	Existing	630	5.8	11.5	☐	1°C/min	0.0	
130		R20	Existing	250	2.3	4.6	☐	1°C/min	0.0	
131		R21	Existing	1600	14.7	29.3	☐	1°C/min	0.0	
132		R22	Existing	630	5.8	11.5	☐	1°C/min	0.0	
133		R25	New Extension	5000	45.8	91.6	☒	1°C/min	91.6	
134		R26	New Extension	8000	73.3	146.5	☒	1°C/min	146.5	
135		R27	New Extension	5000	45.8	91.6	☒	1°C/min	91.6	
136		R28	New Extension	8000	73.3	146.5	☐	1°C/min	0.0	
137		R29	New Extension	2500	22.9	45.8	☐	1°C/min	0.0	
138		R30	New Extension	2500	22.9	45.8	☐	1°C/min	0.0	
139		R31	New Extension	5000	45.8	91.6	☐	1°C/min	0.0	
140		R32	New Extension	5000	45.8	91.6	☐	1°C/min	0.0	
141		R33	New Extension	8000	73.3	146.5	☐	1°C/min	0.0	
142		R34	New Extension	8000	73.3	146.5	☐	1°C/min	0.0	
143										
144										
145										
146		Total Simultaneous Active Heat Load			539	kW				
147		Maximum Chiller Capacity			780	kW				
148		Spare Capacity			241	kW				
149		Capacity Status			WITHIN CAPACITY					
150										
151										
152		Summary								
153		Existing reactors active load			209	kW				
154		New extension reactors active load			330	kW				
155		Total simultaneous load			539	kW				
156		Maximum chiller capacity			780	kW				
157		Spare capacity			241	kW				
158		% utilisation			69	%				
159										

Project: Asymchem PDF Extension Facility Project No: 300291 Subject: Utility Capacity Assessment Calculations By: ME Checked: MC Approved: ME Date: 21-May-26 Document No: 300291-PR-CA-0001 Sheet Name: H2 Cylinders Revision: A1																			
Line No.	Calculation														Notes				
1	Hydrogen cylinders for hydrogenation suite																		
2																			
3	1) To estimate the hydrogen cylinder usage for the new extension facility and new 2500L hydrogenation reactor																		
4	Scenario: Hydrogen cylinders are used to supply the hydrogenation suite and hydrogenation reactors. For the existing facility, a 250L hydrogenation reactor using 1 cylinder per batch and 3 batches are manufactured per week. For the new extension facility, a 2500L hydrogenation reactor will be installed. Scale-up will be applied for the hydrogen cylinder usage.																		
5																			
6	Existing facility																		
7	Hydrogenation reactor volume250L																		
8	Existing facility H2 cylinder usage1H2 cylinders/ batch																		
9	Number of batches per week3batches/ week																		
10	Total H2 cylinders per week3H2 cylinders/ week																		
11	Extension facility																		
12	Hydrogenation reactor volume2500L																		
13	Reactor multiplier factor10																		
14	Existing facility H2 cylinder usage10H2 cylinders/ batch																		
15	Number of batches per week3batches/ week																		
16	Total H2 cylinders per week30H2 cylinders/ week																		
17	It is recommended to utilise hydrogen MCPs (Multi-Cylinder Packs) for H2 cylinder usage																		
18	BOC MCPs house 15 H2 cylinders (108.15m3). To provide enough H2 for the hydrogenation reactors, 2 MCPs are estimated to be used per week at peak usage. It is recommended to have 2 MCPs in the storage location and have 1 MCP available as spare to replace one of the other 2 in use should they deplete																		
19																			
20																			
21	Typical dimensions for each MCP is:																		
22	Length1290mmBetween MCPs800mm																		
23	Width840mmAbove MCPs400mm																		
24	Height1810mm																		
25																			
26	3 MCPs total in storage (2 in use and 1 spare)																		
27	Allowing 500mm distance between MCPs and including 400mm height allowance for distribution manifold, total footprint is:																		
28																			
29	Length5470mmLength3380mm																		
30	Width1680mmfor 3 MCPsWidth1680mmfor 2 MCPs																		
31	Height2210mmHeight2210mm																		
32																			
33	2) Estimating hydrogen storage mass per MCP for COMAH consideration																		
34	Each MCP storage hydrogen at 15C and 175barg. Cylinder gross weight is 1500kg and volume is 108.15m³.																		
35	At these conditions, hydrogen deviates from ideal gas behaviour and behaves as a real gas.																		
36	Per MCP																		
37	MCP storage volumeV108.15m³@ standard conditions (25C, 1atm)																		
38	No. of cylinders per MCP15cylinders																		
39	H2 storage densityρ0.0832kg/m³@ standard conditions (25C, 1atm)																		
40	H2 storage mass per MCPmH29.0kg per MCP																		
41	H2 storage mass per 3 MCPs27.0kg per 3 MCPs54																		
42																			
43																			
44	Summary:																		
45	Recommendation to use Multi-Cylinder Packs (MCPs). BOC MCPs house 15 cylinders each.																		
46	At peak usage, 2 MPCs in use simultaneously, plus 1 spare on standby = 3 MCPs total on site at any time. Footprint for this needs to be accommodated in hazardous storage design. 3rd/ spare MCP may be located elsewhere and brought in during changeover.																		
47																			
48	This is based on batch frequency of 3 batches/week.																		
49	Note - An assessment into the total H2 storage on-site will need to be done to determine H2 storage for COMAH requirements. It is very unlikely the COMAH lower tier of 5t of H2 storage here.																		
50																			
51																			
52																			

Project:Asymchem PDF Extension Facility Project No:300291 Subject:Utility Capacity Assessment Calculations By:ME Approved:ME Checked:MC Date:21-May-26 Document No:300291-PR-CA-0001 Sheet Name:LN2 Tank Revision:A1																	
Line No.	Calculation															Notes	
1	Liquid Nitrogen Tank															Assume 0.5m height for skirt and dished ends	
2																	
3	1) To estimate the nitrogen usage in the new extension facility and whether the existing liquid nitrogen tank fill-up rate can cope with the added requirement from the extension facility																
4																	
5	Scenario: LN2 is used in the existing facility for the cryogenic reactors. Gaseous N2 is used for pressure testing reactors																
6	and for inert blanketing/ boil-off of the reactors. Both are supplied from an existing LN2 storage tank. This also consists of																
7	a nitrogen supply package and evaporator which can supply up to 150Nm3/hr of N2 at 6barg. Currently, the LN2 tank is																
8	refilled every 1-2 weeks with peak demand requiring fill-up every week.																
9	Existing nitrogen supply package can provide 150 Nm3/hr of N2 at 6barg.																
10	Existing LN2 storage tank																
11	Diameter1900mm1.9m																
12	Overall height7480mm7.48m																
13	Tank T-T height (excl skirt)6980mm6.98m																
14	Tank working volume10.00m³																
15	Volume of dished ellipsoidal ends1.8m³																
16	Tank gross volume11.8m³																
17																	
18	Scale-up LN2 storage tank																
19	Scaled-up tank working volume35m³																
20	Scaled-up tank gross volume42m³																
21																	
22																	
23	Summary:																
24	Existing LN2 tank will be insufficient alone to provide required liquid and gaseous N2 for both the existing and																
25	new extension facility with the current tank top-up every week. 2 options are recommended:																
26	1. More frequent top-up of the existing LN2 tank. Currently this is topped up once a week at peak demand. With																
27	the extension facility, this will need to be topped up every 3-4 days.																
28	2. Install a new larger LN2 storage tank based on the scale-up in N2 usage. For new tank installation, footprint will																
29	require new concrete plinth at 6.5m x 10.5m for all tank sizes at 30m3 - 70m3. New tank will need to be minimum																
30	35m3 working volume. Existing tank area is not suitable for larger tank due to condition of existing plinth																
31	(confirmed with BOC). Thus, new tank will need to be located elsewhere then existing tank disinvested once the																
32	new tank is installed and is able to supply both existing and extension facilities.																
33																	

Project:Asymchem PDF Extension Facility Project No:300291 Subject:Utility Capacity Assessment Calculations By:ME Approved:ME Checked:MC Date:21-May-26 Document No:300291-PR-CA-0001 Sheet Name:Compressed Air Revision:A1																
Line No.	Calculation														Notes	
1	Compressed air for plant and instrument air usage														Information on material properties obtained from engineeringtoolbox	
2																
3	1) To estimate the compressed air usage at the existing facility and whether the existing compressors can cope with the increased usage with the extension facility														Estimate each valve at 0.3 Nm3/hr Assume 10 NL/store and 3 strokes/hr 3 existing diaphragm pumps From pilot plant equipment list	
4	Scenario: Compressed air is supplied to the existing facility from on-site compressors. The existing compressed air package supplies 375Nm3/hr @ 10barg to plant air, instrument air and breathing air users. Current breathing air package capacity is 100Nm3/hr. This leaves 275m3/hr available for plant air and instrument air.															
5	Asymchem have provided a valve and instrument list of their existing facility to estimate the CA usage at the current facility. Additionally, there are 3x diaphragm pumps with intermittent use which will also be considered in the CA use.															
6																
7	This will be used to estimate the existing facility CA use and scale it accordingly for the extension facility. The extension facility will have larger equipment whereas the existing facility has more equipment. As a result, a scale-up ratio of 1.5 will be used to estimate the CA usage in the extension facility compared to the existing one.															
8																
9																
10																
11																
12																
13																
14	Based on the valve and instrument list provided, plant air use for existing facility is:															
15	Positioners (continuous bleed)58Nm3/hr															
16	Purges & tools (continuous)10Nm3/hr															
17	On/off valve actuators (intermittent)16Nm3/hr															
18	Diaphragm pumps (intermittent)90Nm3/hr															
19	Peak demand174Nm3/hr															
20																
21	Existing compressor capacity375Nm3/hr @ 10barg															
22	Breathing air package capacity100Nm3/hr															
23																
24	Applying 1.5 scale-up factor, plant air use for both existing and new extension facilities is:															
25	Scale-up factor1.5															
26	Positioners87Nm3/hr															
27	Purges & tools15Nm3/hr															
28	On/off valve actuators (0.5 diversity)12Nm3/hr0.5diversity															
29	Diaphragm pumps (0.7 coincidence)105Nm3/hr0.7diversity															
30																
31	Peak demand across both existing and extension facilities219Nm3/hr															
32																
33	Spare capacity56Nm3/hr															
34	Spare margin15%															
35																
36																
37																
38	Summary:															
39	Existing 375Nm3/hr compressor package has sufficient capacity to provide compressed air to both the existing and extension facilities. The compressed air utilisation for plant air is estimated to be 209Nm3/hr thus, leaving 56Nm3/hr as spare capacity (noting 100Nm3/hr breathing air requirement. An additional compressor would be beneficial to allow larger spare capacity for future upgrades and equipment. However, for the feasibility stage, this assessment determines the existing compressor to be sufficient for the requirement.															
40																
41																
42																
43																
44																
45																

Project: Asymchem PDF Extension Facility Project No: 300291 Subject: Utility Capacity Assessment Calculations By: ME Approved: ME Checked: MC Date: 21-May-26 Document No: 300291-PR-CA-0001 Sheet Name: Breathing Air Revision: A1															Scitech EKILUM									
Line No.	Calculation														Notes									
1	Breathing air for personnel in hazardous areas														200 L/min assumed as worst-case flow req. Assume no more than 10 personnel requiring breathing air at one time across both existing and new extension facilities Information on breathing air supply obtained from breathing and compressed air P&ID SAN-902-007-002-000-00099 Receiver SV set at 10barg Assume 9barg operating pressure Assuming ideal gas behaviour Assume breathing air supply for personnel in hazardous areas is continuous (flexible connection) rather than finite air cylinders									
2																								
3	1) To estimate the breathing air flowrate requirement for personnel handling hazardous materials in the reactor rooms in 30 minutes																							
4	Scenario: Compressed air is supplied to personnel handling hazardous materials for respiratory protection. This is due to potential exposure to toxic chemicals, solvent atmospheres etc. This breathing air is supplied to breathing air points of use in these areas which supplies air to breathing hoods which the personnel must wear to operate in the hazardous areas. For this calculation, it is assumed that personnel will be wearing RPE loose hoods which require 170-200 L/min of breathing air per person. The calculation will estimate the breathing air requirement for 30 minutes.																							
5																								
6																								
7	RPE Loose Breathing Hoods - 170 - 200 L/min																							
8	Breathing air flow per person		200		L/min																			
9	Time requirement		30		min																			
10	Breathing air flow across time frame		6000		L/30 min																			
11	Number of personnel across both facilities		10																					
12	Total breathing air requirement across both facilities		60000		L/ 30 min																			
13			60		m³/30 min																			
14																								
15	2) From this, the maximum number of personnel that can be present in the hazardous rooms has been estimated based on the 200L/min air requirement for breathing hoods for 2 scenarios:																							
16	a) Existing breathing air supply package of 100Nm³/hr																							
17	b) Outage in breathing air supply and available breathing air supply based on existing breathing air receiver																							
18																								
19	a) Existing breathing air supply package of 100Nm³/hr																							
20	Existing breathing air supply package		100		Nm³/hr																			
21	Breathing air flow requirement per person		200		NL/min																			
22			12		Nm³/hr																			
23	Maximum number of personnel that can be in hazardous areas across both existing and new extension facilities		8																					
24																								
25																								
26																								
27	b) Outage in breathing air supply and available breathing air supply based on existing breathing air receiver																							
28	Scenario: In case of an outage in the breathing air supply, the maximum breathing air supply capacity will be based on the breathing air receiver (CMP01-GR1). This has a volume of 11m3 and the receiver has an operating pressure of <10barg. The breathing air header has a supply pressure of 6.5barg. In case of an outage, the breathing air supply will be limited to the difference between the receiver pressure and breathing air header pressure (driving force is pressure difference between receiver and header pressure)																							
29																								
30																								
31																								
32																								
33																								
34	Air receiver volume		11		m³																			
35	Air receiver pressure		9		barg		10.01		bara															
36	Downstream regulator set pressure		6.5		barg		7.51		bara															
37	Pressure difference driving force		2.5		barg																			
38																								
39	Breathing air volume available		27.14		m³																			
40	Personnel in hazardous areas require		200		L/min																			
41			6		m³/30 mins																			
42																								
43	Maximum number of personnel that can be present in hazardous areas in case of breathing air outage		4																					
44																								
45																								
46																								
47	Summary:																							
48	Existing 100 Nm³/hr supply is adequate for up to 8 people simultaneously on the basis of 200L/min RPE loose breathing hoods. If peak hazardous area occupancy exceeds this across both facilities, the breathing air supply will need uprating.																							
49																								
50																								
51	In the event of an outage of breathing air supply, air receiver will be able to provide sufficient usable air for maximum 4 people up to 30 minutes (safe evacuation window).																							
52																								
53																								

Date: 21-May-26

Revision: A1



Line No.	Calculation	Notes
1	Cooling Water for Purified Water Trim Cooling	
2		
3	1) To estimate the cooling water requirement for the extension facility	
4	Situation: Cooling water is used for general cooling within the facility as a utility for heat exchangers. Within the existing	
5	facility, this is supplied via Jaeggli Hybrid cooling towers (893kW) using spray water medium at 32C supply and 26C return. For	
6	the extension facility, cooling water is required for trim cooling of the purified water heat exchanger. This is due to heat	
7	generated from the purified water distribution pump which can heat up the purified water over time. Cooling water will be	
8	supplied to the purified water heat exchanger for trim cooling. This will be from a new chiller designated for this cooling water.	
9	The following assumptions are used for estimating the cooling water requirement and chiller size:	
10	- 10 kW heat gain for purified water based on 25m ³ /hr loop flowrate	
11	- Trim cooling heat exchanger duty = 12 kW (25% design margin)	
12	- Cooling water chiller capacity = 15 kW	
13	This is then used to estimate the chilled water flow requirement based on 10C supply T and 15C return T.	
14		
15	Purified water heat gain in loop 10 kW	
16	Design margin 20%	
17	Trim cooling heat exchanger duty 12.0 kW	
18	Chiller duty 15.0 kW	
19	Cooling water calculation	
20	Supply temperature 10 °C	
21	Return temperature 15 °C	
22	Specific heat capacity 4.184 kJ/kg°C	
23	Water density 1000 kg/m ³	
24		
25	Cooling water flowrate 2581 kg/h	
26	2.58 m ³ /h	
27		
28	Summary:	
29	Install a new dedicated chiller for the cooling water flowrate to the purified water trim cooling heat exchanger. This will	
30	cool down incoming cooling water to the required temperature to then be pumped to the purified water trim cooler.	
31	The new chiller will be sized specifically for the purified water trim cooling requirement. Final installation location to	
32	be confirmed at the project concept stage.	
33		

	Project:Asymchem PDF Extension Facility Project No:300291 Subject:Utility Capacity Assessment Calculations By:MEChecked:MC Approved:MEDate:21-May-26 Document No:300291-PR-CA-0001 Sheet Name:Bulk Solvent Revision:A1 															
Line No.	Calculation															Notes
1	Bulk acetone and methanol solvents															Information on material properties obtained from engineeringtoolbox
2																
3	1) To estimate the usage of acetone and methanol solvents from the bulk solvent tanks and whether the existing tanks and changeover is sufficient with the new extension facility															Info from RFI response Bulk tank volumes estimated from HMI screenshot
4	Scenario: Acetone and methanol solvents are used on-site in the reactors. These are stored in 2 solvent tanks: BT01 (methanol, 60m3) and BT01 (acetone, 40m3). Current solvent usage and refilling is confirmed via RFI as: - Acetone: 4 tankers ordered per year (~23.25 MT per delivery). Deliveries every 3 months - Methanol: 6 tankers ordered per year (~27.25 MT per delivery). Deliveries every 2 months on average Note this is based on 4-6 batches per week															
5																Estimated from HMI screenshot
6																
7	Acetone bulk tank volume40m³															Estimated from HMI screenshot
8	Acetone delivery frequency4times per year (quarterly)															
9	Acetone delivery amount23.25MTAcetone density785kg/m³															Estimated from HMI screenshot
10	30m³															
11																Estimated from HMI screenshot
12	Methanol bulk tank volume60m³															
13	Methanol delivery frequency6times per year (bimonthly)															
14	Methanol delivery amount27.25MTMethanol density753kg/m³															Note - this is based on current solvent refill pattern with existing facility which is based on 4-6 batches per week
15	36m³															
16																
17	Solvent usage with new extension facility is estimated based on reactor volume scale-up factor. This is a ratio in the total reactor volumes between the existing facility and new extension facility to estimate the utility usage with the extension equipment (usage data unknown)															
18																
19																
20																
21	Reactor volume scale-up ratio3.54															
22																
23	Acetone delivery frequency with new extension15times per year i.e. ~ one delievery one month then two deliveries the next month															
24	Methanol delivery frequency with new extension22times per year i.e. ~ two delieveries per month.															
25																
26																
27																
28	Summary:															
29	Methanol tanker 27.25 MT deliveries will need to be 2x deliveries for one month then 1x delievery the following month i.e. 3 deliveries every 2 months. 1 delivery for the existing facility operation and 2x delieveries for the new extension facility. The option for installing an additional bulk methanol tank will not be considered as delivery frequency is dependent on tanker sizes and frequent deliveries will still be required even with an additional bulk solvent tank.															
30																
31																
32																
33																

